Chudley-McCullough Syndrome (CMCS): Comprehensive Disease Characteristics Report

Disease: Chudley-McCullough Syndrome (CMCS/CMS) · **MONDO:** MONDO:0011411 · **Phenotype OMIM:** #604213 **Causal gene: GPSM2** (G protein signaling modulator 2; aka *LGN*, *Pins*), HGNC:29501, gene OMIM *609245, NCBI Gene 29899, Ensembl ENSG00000121957, 1p13.3 Category:** Mendelian, autosomal recessive

> **Summary answer:** Chudley-McCullough syndrome is a rare autosomal-recessive disorder caused by **biallelic loss-of-function variants in GPSM2 (LGN/Pins)**. It is defined by **profound congenital sensorineural hearing loss** combined with a **distinctive, largely static constellation of brain malformations** (partial agenesis of the corpus callosum, frontal polymicrogyria, gray-matter heterotopia, inferior cerebellar dysplasia, ventriculomegaly, and arachnoid/interhemispheric cysts) — yet, strikingly, affected individuals usually have **normal or near-normal cognition**. The unifying mechanism is loss of GPSM2-dependent **mitotic-spindle orientation / asymmetric cell division** (brain) and **stereocilia elongation / hair-bundle polarity** (inner ear). Management is supportive, centered on **cochlear implantation**.

Evidence base: this report is derived from **aggregated disease-level literature** (primary case series, molecular genetics papers, and animal/in-vitro mechanistic studies), not from individual EHR data.

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1. Disease Information

Chudley-McCullough syndrome is a clinically recognizable autosomal-recessive syndrome pairing early-onset/congenital severe-to-profound sensorineural hearing loss with a characteristic brain malformation pattern, in the context of preserved psychomotor/cognitive development

(P 22578326 ; P 41323166). P 33016209

First described in 1997 by Chudley, McCullough and McCullough in a consanguineous Manitoba Mennonite brother–sister pair; the causal gene (GPSM2) was identified in 2012 by Doherty et al. (P 22578326).

Key identifiers: - **MONDO:** MONDO:0011411 - **OMIM (phenotype):** #604213 (CHUDLEY-McCULLOUGH SYNDROME; CMCS) - **OMIM (gene GPSM2):** *609245 - Orphanet: Chudley-McCullough syndrome (ORPHA; rare genetic deafness-malformation entity* — specific ORPHA code not verified in this analysis*) - ICD-10: no dedicated code; component codes apply — H90.3 (bilateral sensorineural hearing loss), Q04.0 (congenital malformations of corpus callosum), Q04.3 (other reduction deformities of brain) - ICD-11: no dedicated stem; map via structural developmental anomalies of the brain + sensorineural hearing loss - MeSH: no unique MeSH descriptor (covered by supplementary concept / indexed under the component terms) - HGNC:** HGNC:29501 (GPSM2)

Synonyms / alternative names: CMCS; CMS; Chudley-McCullough syndrome; (historically allelic designation) **DFNB82** nonsyndromic hearing loss — now understood to lie on the same GPSM2 phenotypic spectrum (P 22578326 ; P 20602914).

2. Etiology

Primary cause — genetic (Mendelian, monogenic): Biallelic (homozygous or compound heterozygous) loss-of-function variants in **GPSM2** (P 22578326).

"*Homozygosity mapping and whole-exome sequencing … revealed four molecular variations (two*

single-base deletions, a nonsense mutation, and a canonical splice-site mutation) in the G protein-signaling modulator 2 gene, GPSM2, that underlie CMS" (22578326).

Genetic risk factors: - **Causal variants:** null alleles in GPSM2 (nonsense, frameshift, canonical splice-site). Examples: p.Ser352 *(c.1055C>A;* *27180139*), *p.Arg365* (c.1093C>T; 27064331), c.1473delG founder frameshift (23494849). - **Susceptibility/modifier loci:** None established. No disease-modifier gene has been identified. - **Family history / consanguinity:** Historically strong — many families are consanguineous or from genetically isolated populations (22578326; 23494849). However, non-consanguineous cases without prior family history occur (38567212).

Environmental risk factors: None known. No toxin, infection, drug, maternal exposure, radiation, age, or sex effect is implicated. It is a purely monogenic developmental disorder.

Protective factors: Not applicable in a classical sense. The only "protective" genetic state is the absence of a second pathogenic GPSM2 allele — **heterozygous carriers are unaffected** (consistent with gnomAD pLI ≈ 0; §4, §9).

Gene–environment interactions: None documented; environment is not a known contributor.

3. Phenotypes

All phenotypes are **congenital/prenatal in origin**, **bilateral**, **non-progressive** (except occasional ventriculomegaly), and **lifelong**.

Phenotype	Type	Onset	Severity	Frequency	HPO
Sensorineural hearing loss (severe–profound, bilateral)	Clinical sign / sensory	Congenital	Severe–profound	~100% (defining)	HP:0000407 (sensorineural hearing impairment); HP:0008527 (congenital SNHL)
Partial agenesis of corpus callosum (esp. splenium)	Structural (imaging)	Prenatal	—	Hallmark, very high	HP:0001338 (partial ACC); HP:0001274 (abnormal CC)
Frontal polymicrogyria	Structural (imaging)	Prenatal	—	"Uniformly present" (P 22578326)	HP:0002126 (polymicrogyria)
Gray-matter heterotopia (frontal subcortical/ periventricular)	Structural (imaging)	Prenatal	—	"Uniformly present"	HP:0002282 (gray matter heterotopia); HP:0030047
Cerebellar dysplasia (inferior)	Structural (imaging)	Prenatal	—	Nearly invariant	HP:0001317 (abnormal cerebellum); HP:0007360
Ventriculomegaly / colpocephaly / slit-like frontal horns	Structural (imaging)	Prenatal	Variable	Nearly invariant (variable degree)	HP:0002119 (ventriculomegaly)
Arachnoid / interhemispheric cyst	Structural (imaging)	Prenatal	Variable	Nearly invariant	HP:0100702 (arachnoid cyst)
Hydrocephalus (subset)	Clinical/ structural	Neonatal	Variable (shunt in some)	Minority	HP:0000238 (hydrocephalus)

Phenotype	Type	Onset	Severity	Frequency	HPO
Balance/vestibular dysfunction (inferred; shown in mouse)	Sign	—	Variable	Uncertain in humans	HP:0002131 / vestibular dysfunction
Preserved cognition / normal development (a defining *negative* feature)	—	—	—	Typical	(notable *absence* of HP:0001249 intellectual disability)

Key quote: "*frontal polymicrogyria and gray matter heterotopia are uniformly present, whereas cerebellar dysplasia, ventriculomegaly, and arachnoid cysts are nearly invariant. Despite these striking brain malformations, individuals with CMS generally do not present with significant neurodevelopmental abnormalities, except for hearing loss*" (P 22578326).

Quality-of-life impact: Dominated by profound deafness (language/communication, education, social participation) — largely remediable with cochlear implantation (§12). Cognitive/motor QoL is generally preserved. No disease-specific validated QoL instrument exists (generic pediatric hearing-loss/EQ-5D/SF-36 tools would apply). Hydrocephalus, when present, adds neurosurgical morbidity.

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4. Genetic / Molecular Information

Causal gene: GPSM2 (HGNC:29501; gene OMIM *609245; Ensembl ENSG00000121957; chr1:108,875,350–108,934,545 GRCh38; cytoband 1p13.3). The protein GPSM2/LGN is a scaffolding cell-polarity protein with N-terminal TPR (tetratricopeptide) repeats (bind mInsc and NuMA) and C-terminal GoLoco/GPR motifs** (bind Gαi-GDP)

(P 11781568 ; P 22074847). P 27180139

Pathogenic variants: - **Type/class:** Predominantly **truncating loss-of-function** — nonsense, frameshift (single-base deletions), and canonical splice-site. No recurrent pathogenic missense class established. Representative alleles: c.1055C>A/p.Ser352 (P *27180139*); *c.1093C>T/p.Arg365* (P 27064331); c.1473delG (Dutch/Mennonite founder frameshift; P 23494849). Truncations "*abolish all four GoLoco domains in GPSM2*" (P 27180139).

- **Classification (ACMG/AMP):** Reported disease alleles are **Pathogenic**, with null mechanism supporting PVS1-type evidence. - **Allele frequency (gnomAD):** Individual pathogenic null alleles are rare. Gene-level constraint: **pLI = 6.4×10^{-15} (≈0)**, **oe_lof = 0.74 (90% CI 0.60–0.92; LOEUF 0.92)**, obs_lof 58 / exp_lof 78.4, lof_z = 1.96 — i.e., GPSM2 is **tolerant of heterozygous LoF**, fully consistent with recessive inheritance (carriers unaffected). - **Somatic vs germline: Germline** only; no somatic/cancer association, no mosaicism required. - **Functional consequence: Loss of function** (not gain-of-function or dominant-negative).

Modifier genes: None identified (intrafamilial variability — e.g., discordant twins — suggests stochastic/epigenetic rather than defined modifiers; P 27064331).

Epigenetic information: No disease-specific DNA-methylation/histone signature reported.

Chromosomal abnormalities: None — CMCS is a single-gene point/indel disorder; CMA/karyotype are normal and low-yield.

5. Environmental Information

- **Environmental factors:** None known (no toxin/radiation/pollution/occupational link).
- **Lifestyle factors:** None relevant.
- **Infectious agents:** None — CMCS is genetic. (Clinically important *differential*: congenital infections such as CMV can mimic SNHL + brain anomalies, so infectious causes must be excluded diagnostically; §10.)

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic GPSM2 (LGN) loss-of-function variants** → truncated protein lacking GoLoco (Gαi-binding) motifs → **loss of a functional LGN/Pins scaffold** (*demonstrated*; P 22578326; P 27180139).
2. Loss of LGN **results in** failure to assemble the cortical **Gαi–LGN–NuMA** spindle-tethering complex at the cell cortex (*demonstrated in vitro/cell models*; P 11781568).
3. **Branch A — Brain:** Failure of the LGN complex **leads to** mis-oriented mitotic spindles / disrupted **asymmetric cell division (planar polarity)** in neural progenitors during fetal corticogenesis (*inferred for human brain from LGN biology*; P 22578326) → disordered neuronal production/migration → **polymicrogyria, subcortical/periventricular gray-matter heterotopia, callosal (splenial) agenesis, cerebellar dysplasia**, and impaired CSF-space/ependymal development → **ventriculomegaly, interhemispheric/arachnoid cysts**.
4. **Branch B — Inner ear:** Loss of apical **Gpsm2–Gαi** at hair-cell stereocilia tips **results in** defective **actin-rich stereocilia elongation** and loss of row-1 tip-complex (Gpsm2-Gαi-Whirlin-Myo15-Eps8) assembly (*demonstrated in mouse & in vitro*;

; P 28387217 ; P 35687681) P 20602914

→ short/disorganized hair bundles → cochlear (and vestibular) hair-cell dysfunction → **profound sensorineural hearing loss** (± balance defects).

5. Because the lesion acts during a **prenatal critical window** and GPSM2 is dispensable for post-mitotic neuronal *function*, the malformations are **static** and **cognition is typically preserved** (*inferred*; P 22578326 ; P 27312216).

Detail by category

- **Molecular pathways / machinery:** G-protein (Gαi) signaling–modulated **mitotic spindle orientation** and **planar cell polarity**; LGN–NuMA–dynein spindle-pulling. Not a classical Wnt/MAPK/mTOR disease. GO:BP: **GO:0000132** (establishment of mitotic spindle orientation), **GO:0007163** (establishment/maintenance of cell polarity), **GO:0001736** (establishment of planar polarity), **GO:0060113** (inner ear receptor cell differentiation), **GO:0007605** (sensory perception of sound).
- **Cellular processes:** Asymmetric cell division, spindle positioning, actin-bundle (stereocilia) morphogenesis. Not apoptosis/inflammation driven.
- **Protein dysfunction:** Truncation → loss of Gαi-binding GoLoco motifs → loss-of-scaffold function (no aggregation/GoF).
- **Metabolic / immune / epigenetic:** No primary metabolic, immune, or epigenetic contribution described.
- **Subcellular compartments (GO:CC): GO:0005938** (cell cortex), **GO:0000922** (spindle pole), **GO:0120044** (stereocilium tip), **GO:0032420** (stereocilium).
- **Cell types (CL):** cochlear/auditory hair cell **CL:0000202** (inner **CL:0000589**, outer **CL:0000601**), vestibular hair cell, inner-ear supporting cell, neural/radial-glial progenitor **CL:0000681**.
- **Molecular profiling:** No human CMCS transcriptomic/proteomic/metabolomic signature published; mechanistic data come from mouse and biochemical/biophysical (phase-

separation) studies
( ).
P 35687681

7. Anatomical Structures Affected

- **Organ/system level — primary: Central nervous system** (UBERON:0001017) and **inner ear / auditory-vestibular system** (UBERON:0001846).
- Corpus callosum (**UBERON:0002336**) — partial agenesis, splenium-predominant
- Frontal cerebral cortex (**UBERON:0016525**) — polymicrogyria
- Subcortical/periventricular white & gray matter — heterotopia
- Cerebellum (**UBERON:0002037**) — inferior dysplasia
- Lateral ventricles (**UBERON:0002285**) — ventriculomegaly/colpocephaly, slit-like frontal horns
- Arachnoid/subarachnoid & interhemispheric space — cysts
- Cochlea (**UBERON:0001844**) and vestibular organs — hair-cell/stereocilia pathology
- **Secondary involvement:** Obstructive hydrocephalus (subset) → neurosurgical shunt.
- **Tissue/cell level:** Neuroepithelium/neural tissue (nervous tissue); inner-ear sensory epithelium (specialized epithelial hair cells). CL:0000202 (hair cell), CL:0000681 (radial glial/progenitor).
- **Subcellular:** cell cortex / mitotic spindle pole (brain); stereocilium tip/actin core (ear). GO:CC as in §6.
- **Localization / lateralization: Bilateral/symmetric** for deafness; brain malformations **bilateral with midline/frontal emphasis**.

8. Temporal Development

- **Onset: Congenital/prenatal.** Brain malformations form during fetal neurogenesis; detectable prenatally on fetal MRI/US (slit-like frontal horns, ACC, frontal subcortical heterotopia, arachnoid cysts, cerebellar dysplasia;

). P 27312216
Hearing loss present at/from birth.

- **Onset pattern:** Insidious/developmental (structural), not acute.
- **Progression: Static/non-progressive** malformations; cognition stable. **Exception:** ventriculomegaly can progress to hydrocephalus in a subset (neonatal), occasionally requiring shunt
( ). P 27064331
- **Course/duration:** Chronic, lifelong, non-degenerative.
- **Remission:** None spontaneously; hearing is **functionally restorable** by cochlear implant (treatment-induced functional "remission" of the deafness).
- **Critical period:** The **prenatal window of progenitor division/hair-cell differentiation** is the vulnerability window; postnatally the key intervention window is **early cochlear implantation** for language development.

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9. Inheritance and Population

- **Inheritance: Autosomal recessive** (OMIM #604213).
- **Penetrance:** Effectively **complete** for profound congenital SNHL and core cortical malformations in biallelic-null individuals.
- **Expressivity: Variable** for ventriculomegaly/hydrocephalus and cyst burden — even **discordant between twins** with identical genotype
( ); P 27064331
some GPSM2 individuals (originally "nonsyndromic"/DFNB82) have cortical malformation but **no ventriculomegaly**
( ). P 22578326
- **Anticipation:** None (not a repeat-expansion disorder).
- **Germline mosaicism:** Not specifically reported.
- **Founder effects: Dutch/Mennonite c.1473delG** founder haplotype traced to Europe and carried to North America

(P 23494849);
Palestinian DFNB82 founder
(P 20602914).
Additional families: Yemeni
(P 27180139),
others.

- **Consanguinity:** Major contributor historically (homozygous null in consanguineous/isolated populations), but non-consanguineous compound-heterozygous cases exist (P 38567212).
- **Carrier frequency:** Not formally estimated; gnomAD LoF alleles individually rare; elevated locally in founder populations.
- **Epidemiology: Ultra-rare** — reported **~30 times** in the literature since 1997 (P 38567212).
  No validated prevalence/incidence figures (too rare for registry estimates).
- **Demographics:** Panethnic reports; clustering in consanguineous/founder populations. **Sex ratio ≈ 1:1** (expected for AR; no sex bias). Age distribution: diagnosed in infancy/childhood (congenital), individuals survive into adulthood.

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10. Diagnostics

- **Imaging (cornerstone): Brain MRI** showing the recognizable pattern — partial ACC (splenium), frontal polymicrogyria, frontal subcortical/periventricular heterotopia, inferior cerebellar dysplasia, ventriculomegaly/colpocephaly with slit-like frontal horns, and arachnoid/interhemispheric cysts (P 22578326 ; P 27312216 ; P 41323166).
  Prenatal fetal imaging can suggest the diagnosis and is prognostically important

(P 27312216).
RadLex/NCIT: MRI (NCIT:C16809).

- **Audiology:** Newborn hearing screen fail; ABR/OAE/audiometry confirm **severe–profound bilateral SNHL**. LOINC-coded audiometry/ABR.
- **Genetic testing (confirmatory):** Detection of **biallelic GPSM2 pathogenic variants** via **single-gene testing**, **hereditary-hearing-loss gene panels**, or **whole-exome/whole-genome sequencing** (WES/WGS). WES + homozygosity mapping was the original discovery route
(P 22578326 ; P 20602914 ; P 23494849).
An **LOVD GPSM2 database** supports interpretation
(P 23494849).
GTR: GPSM2 single-gene and deafness-panel tests. NCIT: Genetic Testing (NCIT:C15709).
- **Low-yield/not indicated:** Chromosomal microarray, karyotype, FISH, mtDNA, repeat-expansion testing (no structural/mitochondrial/repeat basis).
- **Laboratory/biomarkers:** No specific biochemical blood/urine biomarker; genotype is the molecular marker.
- **Omics diagnostics:** Not used clinically (no RNA/proteomic/metabolomic diagnostic assay).
- **Clinical criteria:** No formal consensus criteria; diagnosis is **pattern recognition (SNHL + characteristic MRI) + GPSM2 confirmation**.
- **Differential diagnosis:** Other malformations of cortical development / ACC syndromes; **congenital CMV**; tubulinopathies; Aicardi syndrome; other syndromic deafness. **Distinguishing feature: preserved cognition despite severe imaging findings**
(P 27312216)
— key for counseling because prognosis is far better than look-alikes.
- **Screening:** Captured indirectly by **newborn hearing screening** (triggers evaluation); **carrier/cascade testing** in families and founder populations.

11. Outcome / Prognosis

- **Survival/mortality: Life expectancy appears normal**; no disease-specific mortality described. CMCS is non-degenerative.
- **Morbidity/disability:** Dominated by **profound deafness** → communication/language disability if untreated. Cognition and motor function generally preserved; thus overall functional prognosis is **favorable**, markedly better than morphologically similar brain-malformation syndromes (P 27312216).
- **Complications:** Hydrocephalus requiring shunt in a subset (P 27064331); shunt-related complications possible. Deafness-related developmental/educational impact if not remediated.
- **Recovery potential:** Brain malformations are fixed/non-recoverable but non-progressive; **hearing is functionally recoverable with cochlear implantation** (§12).
- **Prognostic factors:** Timeliness of hearing habilitation (early CI → better language outcomes); presence/progression of ventriculomegaly/hydrocephalus. No molecular prognostic biomarker beyond genotype (all null → consistent core phenotype).
- **QoL measures:** Generic pediatric hearing-loss/EQ-5D/SF-36/PROMIS; no CMCS-specific tool.

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12. Treatment

No disease-modifying, pharmacologic, gene, cell, or RNA therapy exists. Management is **supportive and multidisciplinary**, targeting the deafness and any hydrocephalus.

- **Hearing habilitation (primary): Bilateral cochlear implantation** effectively restores functional hearing — "*two successful cases of bilateral cochlear implantation in two children with CMS*" (P 38567212); additional CI reports (P 30907716 ;

Forli 2019). Hearing aids/auditory-verbal therapy as adjuncts. NCIT: **Cochlear Implantation (NCIT:C53473)**.

- **Neurosurgical: Ventriculoperitoneal shunt** for hydrocephalus when present/progressive (P 27064331). NCIT: shunt/ventriculoperitoneal shunt procedure.
- **Supportive/rehabilitative:** Speech-language therapy, early-intervention/educational support, audiologic follow-up; physical/occupational therapy if balance/vestibular issues. NCIT: Supportive Care (NCIT:C15399), Speech Therapy.
- **Pharmacotherapy / pharmacogenomics:** None specific.
- **Experimental / clinical trials:** No CMCS-specific trials identified; GPSM2 is not currently a gene-therapy target.
- **Treatment outcomes:** CI outcomes reported as successful hearing restoration (P 38567212); no CMCS-specific adverse-event profile beyond standard CI/shunt risks.
- **Personalized medicine:** Diagnosis-guided counseling; otherwise genotype does not alter management (uniform null mechanism).

13. Prevention

- **Primary prevention:** Not preventable at the individual level (congenital genetic). Population-level risk reduction via **genetic counseling** and reduced consanguinity in high-risk groups.
- **Genetic counseling (central):** AR recurrence risk **25%** for carrier couples; cascade carrier testing; discussion of options. NSGC/ACMG/GeneReviews frameworks apply.
- **Reproductive options / secondary prevention: Preimplantation genetic testing (PGT-M)** and **prenatal diagnosis** (molecular if familial variants known; fetal MRI pattern recognition otherwise) — prenatal diagnosis is feasible and prognostically valuable (P 27312216).
- **Carrier/founder screening:** Targeted carrier screening in founder populations (Dutch/Mennonite c.1473delG; Palestinian DFNB82)

(P 23494849 ; P 20602914).

- **Newborn screening:** Universal **newborn hearing screening** enables early detection and timely CI (secondary prevention of language disability).
- **Tertiary prevention:** Early cochlear implantation; monitoring/treatment of hydrocephalus.
- **Immunization / behavioral / environmental / prophylaxis:** Not applicable (non-infectious, non-environmental).

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14. Other Species / Natural Disease

- **Orthologous gene:** Mouse **Gpsm2** (NCBI Gene 76123); the gene/pathway (LGN/Pins) is **deeply conserved** — *Drosophila* **Pins**, mammalian **LGN/AGS3** paralogs (P 11781568 ; P 22074847 ; P 37017303).
- **Species affected (experimental):** *Mus musculus* (**NCBITaxon:10090**). No naturally occurring companion-animal/wildlife CMCS-equivalent disease is cataloged in OMIA (not reported here).
- **Comparative biology:** The spindle-orientation (Pins/LGN–NuMA/Mud–Gαi) module is conserved from *Drosophila* neuroblasts to mammalian progenitors and epithelia (P 11781568 ; P 37017303), underpinning cross-species mechanistic study. Paralog **AGS3/Gpsm1** modulates LGN function (P 37017303 ; P 28684399).
- **Zoonotic potential / transmission:** Not applicable (genetic, non-transmissible).

15. Model Organisms

- **Mouse (primary model):** *Gpsm2* mutant mice recapitulate the **auditory phenotype** and reveal **vestibular/balance defects** via impaired **actin-rich stereocilia elongation** — "*mutation of the mouse orthologue of GPSM2 affects actin-rich stereocilia elongation in auditory and vestibular hair cells, causing deafness and balance defects*" (P 28387217 ; correction P 29799026). GPSM2 localizes to the apical surface of hair/supporting cells, highest in embryonic development (P 20602914).
- **In-vitro/biochemical models:** Reconstitution showing Gpsm2-Gαi **phase separation** drives row-1 tip-complex (5xTCD: Gpsm2-Gαi-Whirlin-Myo15-Eps8) condensate formation and actin bundling; a deafness-associated Gpsm2 mutation impairs this — "*A deafness-associated mutation of Gpsm2 leads to impaired formation of the 5xTCD condensates and consequently reduced actin bundling, providing possible clues for etiology of hearing loss in patients with Chudley-McCullough syndrome*" (P 35687681). Crystal structures of LGN–mInsc / LGN–NuMA interfaces inform the spindle mechanism (P 22074847 ; P 11781568).
- **Model types available:** Knockout/mutant mice; cellular spindle-orientation assays; invertebrate (Drosophila Pins) genetics for conserved mechanism.
- **Phenotype recapitulation: Strong for the hearing-loss/stereocilia component**; the **human brain-malformation component is less fully modeled** in published auditory-focused mouse studies (a current limitation).
- **Applications:** Dissecting stereocilia morphogenesis, hair-bundle polarity, and spindle-orientation/asymmetric-division biology; platform for future hearing-loss therapeutics.

• **Resources:** MGI (Gpsm2), IMPC/IMSR, Alliance of Genome Resources.

Supported vs. Refuted Hypotheses

Supported: - GPSM2 biallelic LoF is the cause of CMCS (P 22578326). ✓ - Mechanism = spindle-orientation/asymmetric-division defect (brain) + stereocilia-elongation defect (ear) (P 22578326; P 28387217; P 35687681). ✓ - AR inheritance with LoF-tolerant gene (carriers unaffected; gnomAD pLI≈0). ✓ - Deafness + partial ACC with preserved cognition is a recognizable, prognostically favorable entity (P 22578326; P 27312216). ✓ - Cochlear implantation restores hearing (P 38567212). ✓

Refuted / Not supported: - Gain-of-function or dominant mechanism — refuted (all alleles null; gene LoF-tolerant). - Environmental/infectious causation — not supported (purely genetic). - Strict genotype–phenotype correlation for brain severity — refuted (discordant twins; variable ventriculomegaly; P 27064331; P 22578326).

Limitations & Future Directions

- No formal prevalence/incidence or large natural-history cohort (ultra-rare).
- Human brain-malformation mechanism is **inferred** from LGN cell biology rather than directly demonstrated in human/CMCS neural tissue; brain-specific animal models are limited.
- Modifiers of expressivity (e.g., why some develop hydrocephalus) are unknown.
- No human omics profiling; no targeted/gene therapy.

Key References (PMID)

22578326 (Doherty 2012, landmark gene discovery) · 20602914 (Walsh 2010, DFNB82/GPSM2) · 23494849 (Almomani 2013, founder variant) · 27180139 (Hamzeh 2016, p.Ser352*) · 27064331 (Koenigstein 2016, discordant twins) · 27312216 (Chapman 2016, prenatal diagnosis) · 38567212 (Alter 2024, cochlear implants, ~30 cases) · 33016209 (Blauen 2021, clinical review) · 41323166 (Ouqlani 2026, case/MRI) · 28387217 & 29799026 (Mauriac 2017, mouse) · 35687681 (Shi 2022, tip-complex condensates) · 11781568 (Du 2001, LGN–NuMA) · 22074847 (Yuzawa 2011, LGN–mInsc structure) · 37017303 (Descovich 2023, AGS3–LGN).